Document Control

- System Name: Application
- Version: 1.0
- Author: Data Engineering Team
- Date: 2024-06-12
- Status: Draft

Executive Summary

- The Application facilitates an automated ETL pipeline involving source CSV data ingestion, transformation, cleaning, aggregation, and SCD Type 2 history management, implemented using Delta Live Tables.

Business Objectives

- Automate data ingestion from CSV files to Delta tables
- Ensure data quality through null and duplicate removal
- Aggregate customer spend and maintain accurate, auditable transaction history
- Enable change data capture via SCD Type 2 implementation

In-Scope

- Source CSV data reading and delta table loading
- Schema enforcement for customer and order tables
- Data transformation and cleaning logic
- Creation of ordersummary and customeraggregatespend tables
- Aggregation logic on ordersummary data
- SCD Type 2 implementation for ordersummary
- Process deployment using Delta Live Tables

Out-of-Scope

- Any manual data manipulation
- External system integrations beyond defined delta tables
- Real-time or streaming ingestion
- Data visualization or reporting dashboards

Stakeholders

- Data Engineering Team
- Business Analysts
- Compliance Team

Business Requirements

BR-ETL-001 — Load Source Customer Data

- Description: Read customer CSV data and load to delta table /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata
- Business Value: Automated, structured customer data ingestion
- Priority: High

BR-ETL-002 — Load Source Order Data

- Description: Read order CSV data and load to delta table /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata
- Business Value: Automated, structured order data ingestion
- Priority: High

BR-ETL-003 — Customer Table Schema Enforcement

- Description: Enforce schema with fields CustId, Name, EmailId, Region for customer table
- Business Value: Consistency and data quality assurance
- Priority: High

BR-ETL-004 — Order Table Schema Enforcement

- Description: Enforce schema with fields OrderId, ItemName, PricePerUnit, Qty, Date, CustId for order table
- Business Value: Consistency and data quality assurance
- Priority: High

BR-ETL-005 — Calculate TotalAmount Field

- Description: Add column TotalAmount to order table (PricePerUnit * Qty)
- Business Value: Simplifies downstream aggregation and analytics
- Priority: Medium

BR-ETL-006 — Data Quality: Null and Duplicate Removal

- Description: Remove nulls and duplicates from customer and order tables
- Business Value: Improves analytical accuracy and reliability
- Priority: High

BR-ETL-007 — Create Ordersummary Table

- Description: Create ordersummary table in catalog=gen_ai_poc_databrickscoe, schema=sdlc_wizard if not exists
- Business Value: Centralized storage for enriched order data
- Priority: High

BR-ETL-008 — SCD Type 2 on Ordersummary

- Description: Apply SCD Type 2 methodology to ordersummary based on updates in customer table
- Business Value: Ensures historical accuracy and traceability of data changes
- Priority: High

BR-ETL-009 — Create CustomerAggregateSpend Table

- Description: Create customeraggregatespend table with columns Name, TotalAmount, Date if not exists
- Business Value: Enables spend analysis and reporting
- Priority: Medium

BR-ETL-010 — Aggregate Customer Spend

- Description: Aggregate TotalAmount grouped by Name and Date from ordersummary and load into customeraggregatespend
- Business Value: Facilitates spend tracking for business insights
- Priority: Medium

BR-ETL-011 — Delta Live Tables Implementation

- Description: Implement the entire processing steps using Delta Live Tables
- Business Value: Simplifies workflow orchestration and monitoring
- Priority: High

Assumptions & Constraints

- Source data is available and accessible via defined CSV file paths
- Only the specified schemas will be enforced
- Delta Lake infrastructure is operational

Success Metrics

- Successful, automated runs from CSV to target tables
- Zero null/duplicate records post-processing
- Accurate aggregation results in customeraggregatespend
- SCD Type 2 audit trails present

Risks & Mitigations

- Source data format changes — Mitigation: Schema enforcement and validation
- Data quality issues from source — Mitigation: Robust null and duplicate handling logic
- Infrastructure/permission errors — Mitigation: Pre-execution checks and monitoring

Approval

- Business Owner:
- Data Engineering Lead:
- Compliance Review:
- Date:

Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
customerdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	CSV	Customer records source
orderdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	CSV	Order records source

Appendix B: Target Definitions

	Columns
mmary	OrderId, ItemName, PricePerUnit, Qty, Date, CustId, Name, EmailId, Region, TotalAmount, ValidFrom, ValidTo, IsActi
eraggregatespend	Name, TotalAmount, Date

Appendix C: Transformations

Step	Transformation	Input	Output	Logic
1	Schema enforcement	customerdata, orderdata	as-is tables	Enforce specified column structure
2	Data cleaning	customerdata, orderdata	deduped/clean tables	Remove nulls and duplicate records
3	Calculate TotalAmount	orderdata	orderdata (enriched)	PricePerUnit * Qty
4	Enrich ordersummary	customer and order tables	ordersummary	Join on CustId, include SCD2 fields
5	Aggregate spend	ordersummary	customeraggregatespend	Group by Name and Date, sum TotalAmount

Appendix D: Business Rules

Rule ID	Area	Description	Source
BR-ETL-003	Data	Customer table schema: CustId, Name, EmailId, Region	Requirements
BR-ETL-004	Data	Order table schema: OrderId, ItemName, PricePerUnit, Qty, Date, CustId	Requirements
BR-ETL-006	Data Quality	Remove nulls and duplicates	Requirements

BR-ETL-008	SCD	Apply SCD Type 2 to ordersummary	Requirements

Appendix E: Process Flow

Step	Description	Dependency
1	Read source CSVs for customer and order data	None
2	Enforce schemas for both tables	Step 1
3	Clean data by removing nulls and duplicates	Step 2
4	Add calculated TotalAmount to order table	Step 3
5	Join customer and order data to create ordersummary (with SCD 2 logic)	Step 4
6	Aggregate customer spend from ordersummary	Step 5
7	Write final outputs to target Delta tables	Step 6

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